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Computer Vision
Slot -C

INDUSTRY PROBLEM SOLVING TASK

1. Autonomous Vehicle Vision System Selection

Problem Statement

An automobile company is developing an autonomous vehicle detection system. The system must detect objects such as vehicles, pedestrians, traffic signs, and other obstacles accurately and quickly.

The system has three important constraints:

1. Accuracy must be at least 90%. If accuracy is below 90%, the system fails.
2. Latency must not exceed 70 ms. If latency is more than 70 ms, the response is too slow.
3. The vehicle has moderate computational resources, so a method requiring very high computation is not suitable.

The available methods are:

Method	Accuracy	Latency	Computation
Viola-Jones	82%	30 ms	Low
YOLO	93%	60 ms	Moderate
Deep Learning	97%	90 ms	High

Step 1: Analyze Viola-Jones

Viola-Jones provides 82% accuracy and 30 ms latency with low computational requirements. Its main advantage is its very low latency. It can process images quickly and does not require powerful hardware.

However, the required accuracy is at least 90%, while Viola-Jones provides only 82%.

Therefore:

- **Accuracy requirement:** ❌ 82% < 90%
- **Latency requirement:** ✅ 30 ms < 70 ms
- **Computation requirement:** ✅ Low computation

Although Viola-Jones satisfies the latency and computational requirements, it fails the accuracy requirement.

Since the problem clearly states that the system fails if accuracy is below 90%, Viola-Jones cannot be selected.

Step 2: Analyze YOLO

YOLO provides 93% accuracy, 60 ms latency, and requires moderate computation.

Checking each condition:

- **Accuracy requirement:** ✅ 93% ≥ 90%
- **Latency requirement:** ✅ 60 ms ≤ 70 ms

Computational requirement: ✅ Moderate computation is supported

Therefore, YOLO satisfies all three constraints.

YOLO provides sufficient accuracy to detect objects reliably while maintaining a latency of only 60 ms. Since the vehicle supports moderate computational resources, YOLO can also be deployed without requiring extremely powerful hardware.

Step 3: Analyze Deep Learning

The Deep Learning method provides the highest accuracy of 97%. However, it has a latency of 90 ms and requires high computational resources.

Checking the requirements:

- **Accuracy requirement:** ✅ 97% ≥ 90%
- **Latency requirement:** ❌ 90 ms > 70 ms
- **Computational requirement:** ❌ High computation, while the vehicle supports only moderate computation

Although Deep Learning has excellent accuracy, it fails the latency requirement and also exceeds the available computational capability.

In an autonomous vehicle, latency is extremely important because the vehicle needs to respond quickly to obstacles. A 90 ms processing delay may reduce the time available for braking or changing direction.

Therefore, Deep Learning cannot be selected under the given constraints.

Overall Comparison

Method	**Accuracy ≥ 90%?**	**Latency ≤ 70 ms?**	**Moderate Computation?**	**Result**
Viola-Jones	❌	✅	✅	Reject
YOLO	✅	✅	✅	Select
Deep Learning	✅	❌	❌	Reject

Final Decision

YOLO should be selected for the autonomous vehicle vision system.

The reason is that YOLO is the only method that satisfies all the given constraints simultaneously. It provides 93% accuracy, which is above the minimum required 90%, and has 60 ms latency, which is below the maximum acceptable 70 ms. It also requires moderate computation, which matches the vehicle's available hardware.

Viola-Jones is faster and cheaper computationally, but its 82% accuracy is not sufficient for an autonomous vehicle. Deep Learning has the highest accuracy, but its 90 ms latency is too high and its computational requirement exceeds the available resources.

Conclusion

Therefore, YOLO is the most practical and suitable choice. It provides a good balance between accuracy, response time, and computational requirements. For an autonomous vehicle, this balance is more important than choosing the method with the highest accuracy alone.

Q2. Smart Factory Product Inspection Selection

Problem Statement

A manufacturing company wants to implement a real-time computer vision system to inspect products on a production line. The system must identify defective products quickly and accurately.

The system has three major constraints:

1. Accuracy must be at least 95%. If accuracy is below 95%, defective products may be missed.
2. Processing time must not exceed 50 ms. Higher processing time can slow down the production line.
3. The factory hardware supports moderate computation.

The available methods are:

Method	Accuracy	Processing Time	Computation
Traditional Image Analysis	91%	25 ms	Low
YOLO	96%	45 ms	Moderate
Deep Learning	98%	75 ms	High

Step 1: Analyze Traditional Image Analysis

Traditional Image Analysis provides 91% accuracy, 25 ms processing time, and requires low computation.

Checking the conditions:

- **Accuracy requirement:** ❌ 91% < 95%
- **Processing time requirement:** ✅ 25 ms < 50 ms
- **Computational requirement:** ✅ Low computation

The method is fast and computationally efficient. However, its accuracy is only 91%.
Since the system requires at least 95% accuracy, Traditional Image Analysis cannot be selected.
This is particularly important in a manufacturing environment because a 91% accuracy could result in defective products being classified as good products. Such missed defects can affect product quality and customer satisfaction.
Therefore, Traditional Image Analysis is rejected

Step 2: Analyze YOLO

YOLO provides 96% accuracy, 45 ms processing time, and requires moderate computation.

Checking all the requirements:

- **Accuracy requirement:** ✅ 96% ≥ 95%
- **Processing time requirement:** ✅ 45 ms ≤ 50 ms
- **Computational requirement:** ✅ Moderate computation is supported

YOLO satisfies all the constraints.
Its accuracy is 96%, which is above the required 95%. Its processing time is 45 ms, which is within the maximum acceptable limit of 50 ms.
Since the factory hardware supports moderate computation, YOLO is also suitable from a hardware perspective.

Step 3: Analyze Deep Learning

Deep Learning provides the highest accuracy of 98%. However, its processing time is 75 ms, and it requires high computational resources.

Checking the requirements:

- **Accuracy requirement:** ✅ 98% ≥ 95%
- **Processing time requirement:** ❌ 75 ms > 50 ms
- **Computational requirement:** ❌ High computation

Although Deep Learning has the highest accuracy, it cannot be selected because it takes too long to process each image.

In a production line, processing time is very important. If the system takes 75 ms when the maximum acceptable time is 50 ms, the inspection process may become a bottleneck. This could slow down the production line.

Therefore, Deep Learning is rejected under the given constraints.

Overall Comparison

Method	**Accuracy ≥ 95%?**	**Processing Time ≤ 50 ms?**	**Moderate Computation?**	**Result**
Traditional Image Analysis	❌	✅	✅	Reject
YOLO	✅	✅	✅	Select
Deep Learning	✅	❌	❌	Reject

Final Decision

YOLO should be selected for the Smart Factory Product Inspection System.

It is the only method that satisfies all three requirements simultaneously.

Requirement	YOLO Result	Status
Minimum accuracy	95%	
YOLO accuracy	96%	✅
Maximum processing time	50 ms	
YOLO processing time	45 ms	✅
Available computation	Moderate	
YOLO computation	Moderate	✅

Final Conclusion

YOLO is the best choice because it provides the required accuracy, fast processing, and suitable computational requirements.

Traditional Image Analysis is faster and cheaper but does not provide sufficient accuracy. Deep Learning provides the highest accuracy, but its 75 ms processing time exceeds the 50 ms limit and it requires high computational resources.

Therefore, YOLO gives the best practical balance between accuracy, speed, and hardware requirements and is the most suitable method for deployment in the smart factory.